

Literature Benchmark Brief: Park_2019_Energy

Title: Physics-induced graph neural network: An application to wind-farm power estimation

Authors: J. Park, J. Park (2019)

Venue: Energy | **DOI:** 10.1016/j.energy.2019.115883

Cluster: Cluster A: Spatio-Temporal Wind/Wake

Core Theoretical Mechanism & Model Formulation

Jensen wake decay model embedded as edge weights in graph convolutional network

Empirical Scale & Experimental Dataset

Simulated and operational wind turbine clusters

Relevance to IEEE TSTE & Sustainable Energy Systems

Establishes physics-guided edge adjacency matrices based on wind direction and wake expansion

Identified Gaps & Operational Vulnerabilities

Assumes static offline wind direction, no dynamic time-varying topology update

Role in Manuscript Defense & Reviewer #2 Proof

Justifies dynamic directed wake graph formulation and wake-cone filtering in Section II